

COSC726 AGENTIC AI — PHASE 0: PROJECT INITIATION & ARCHITECTURAL INCEPTION BLUEPRINT

Phase 0: Multi-Agent AI System for Training Course Design & Evaluation

Engineering Vision, Bloom Pedagogical Framework, Multi-Agent Architecture & 8-Phase Planned Roadmap

Project Designer: Mohammad Abulgasim	Supervisor: Dr. Fakhreldeen Saeed
Scope: <i>Phase 0 (Pre-Implementation Architectural Blueprint)</i>	Target Lifecycle: <i>8 Integrated Modular Phases (Phases 1 → 8)</i>

1. Executive Motivation & Problem Formulation:

Standard applications of Large Language Models (LLMs) in educational curriculum design and student evaluation suffer from critical architectural bottlenecks:

1. Arithmetic Hallucinations: LLMs frequently make calculation errors when summing points, computing percentages, and determining GPAs.
2. Lack of Structured Pedagogical Alignment: Monolithic prompts generate unstructured content that lacks alignment with Bloom's Revised Taxonomy.
3. Absence of Objective Traceability: Traditional generative pipelines fail to enforce strict mapping between assessment questions and specific learning objectives.
4. Statelessness: LLMs cannot maintain persistent memory across cohort evaluation lifecycles without ACID-compliant relational databases.
5. Inability to Provide Adaptive Remediation: Generic feedback lacks individualized remedial pathways with scaffolded hints for struggling learners.

This project resolves these challenges by introducing a specialized, coordinated Multi-Agent AI System governed by deterministic validation gates.

2. Strategic Objectives & Pedagogical Framework:

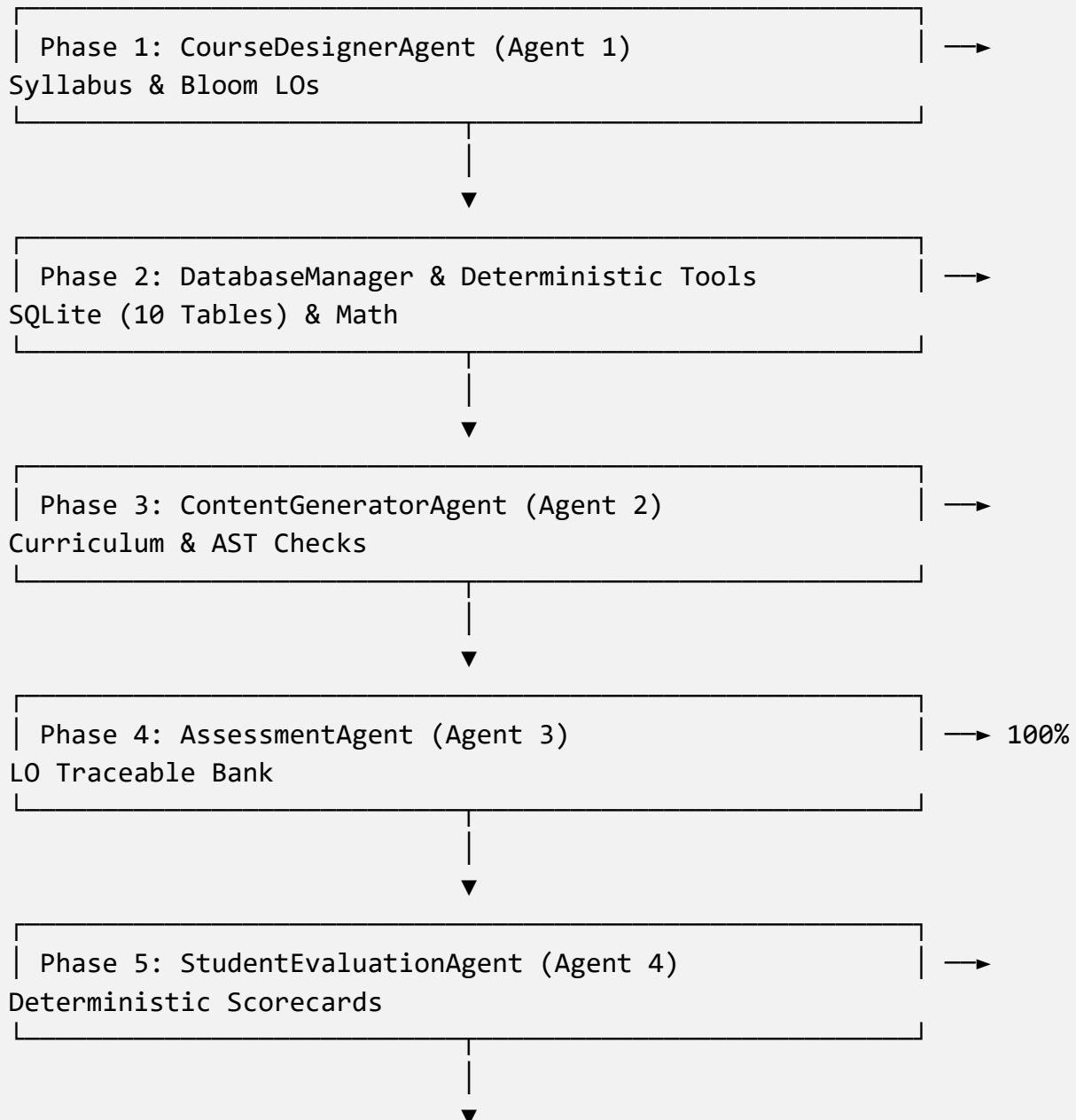
The project is structured around key engineering and pedagogical goals:

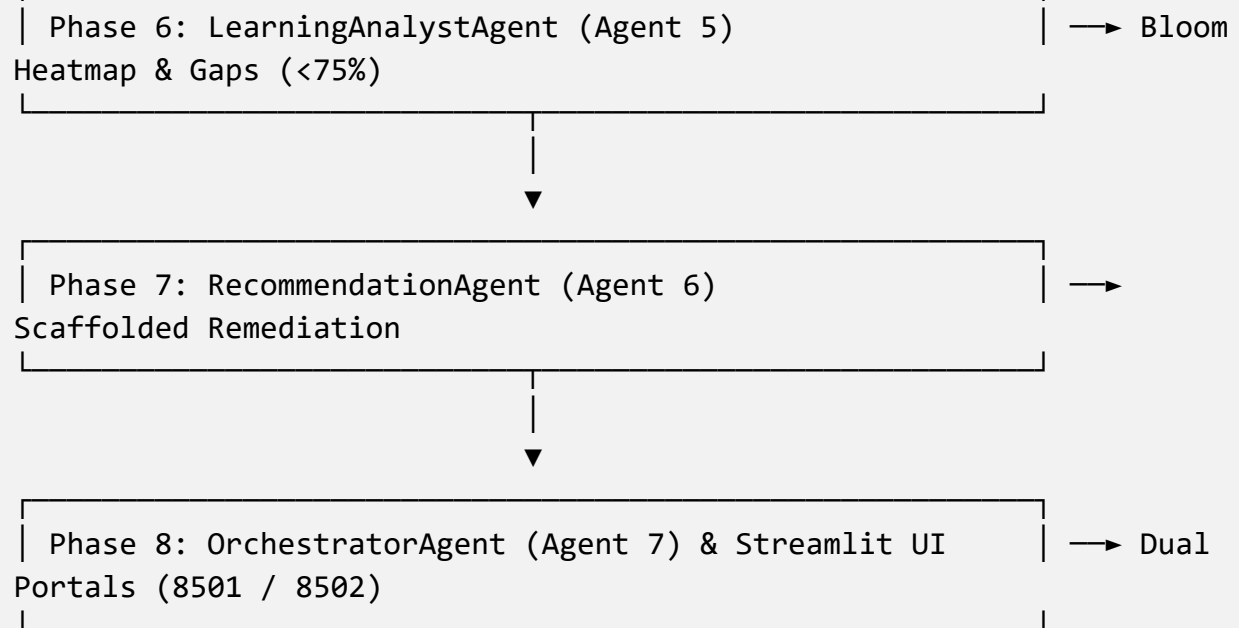
- Autonomous Curriculum Synthesis: Transforming course requirements into structured CoursePlans with balanced hour distribution.
- Full Bloom's Taxonomy Alignment: Mapping objectives across all 6 cognitive tiers (Remember, Understand, Apply, Analyze, Evaluate, Create).
- ACID-Compliant Relational Persistence: Managing 10 SQLite relational tables for courses, questions, submissions, and grades.
- AST Compilation Verification: Ensuring 100% syntax validity of code snippets using Python's Abstract Syntax Tree parser (ast.parse).
- Zero-Hallucination Deterministic Grading: Employing pure mathematical tools for scoring, letter grades, and 4.0 GPA conversion.

- Cohort Heatmaps & Gap Diagnostics: Dynamically diagnosing critical learning gaps where cohort mastery falls below 75%.
- Adaptive Remedial Intervention: Synthesizing customized remediation plans with progressive 3-tier scaffolded hints.
- Isolated Dual-Portal Web Architecture: Providing dedicated spaces for Instructors (Port 8501) and Students (Port 8502).

3. Proposed Multi-Agent Architecture & Pipeline Flow:

End-to-End Planned Multi-Agent Pipeline:





4. The 8-Phase Implementation Roadmap & Target Deliverables:

Planned Phase	Responsible Agent	Primary Function & Scope	Target Deliverables
Phase 1: Course Design	CourseDesignerAgent	Designs syllabus, module hour allocation, and 6-level Bloom objectives with validation gates.	course_plan.json, course_plan.md
Phase 2: Database Persistence	DatabaseManager & Tools	Initializes 10 SQLite relational tables and provides deterministic grading tools.	training_system.db, GradeCalculator
Phase 3: Content Generation	ContentGeneratorAgent	Authors lectures, practical labs, and enforces AST syntax compilation checking.	curriculum_book.md, packages.json
Phase 4: Assessment Bank	AssessmentAgent	Authors 3 diagnostic questions per LO (MCQs & T/F) with 100% hard-coded LO traceability.	assessment_bank.db, suites.json

Phase 5: Student Evaluation	StudentEvaluationAgent	Evaluates submissions with deterministic precision and generates individual scorecards.	student_evaluations.db, scorecards.md
Phase 6: Learning Analytics	LearningAnalystAgent	Aggregates cohort mastery, computes Bloom heatmap, and diagnoses critical gaps (<75%).	learning_analytics.db, heatmap.md
Phase 7: Recommendation Engine	RecommendationAgent	Synthesizes personalized remedial pathways with scaffolded hints and advanced challenges.	recommendation_engine.db, plans.md
Phase 8: Orchestrator & Portals	OrchestratorAgent & UI	Coordinates end-to-end state flow, isolates portals (8501/8502), and provides batch scripts.	unified_platform.db, instructor/student apps

5. Target Quality Gates & Verification Standards:

To ensure uncompromising engineering and pedagogical excellence, the platform establishes 6 mandatory quality gates:

- Gate 1: Strict Course Hour Balance & 6 Bloom Levels Validation Gate.
- Gate 2: AST Python Syntax Compilation Gate (``ast.parse()``) guaranteeing 100% valid code.
- Gate 3: 100% Hard-Coded Learning Objective Traceability Gate.
- Gate 4: Zero-Hallucination Deterministic Mathematical Grading Gate.
- Gate 5: Critical Gap Diagnostic Threshold Gate (< 75.0% Cohort Mastery Alert).
- Gate 6: Strict Architectural Isolation Gate between Instructor (8501) and Student (8502) Portals.